

Name: _____ Date: _____ Period: _____

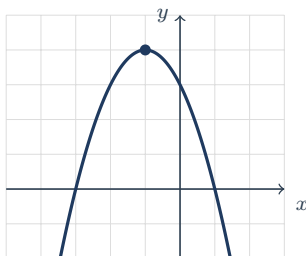
Instructions: Show all work. Give exact answers — simplest radical form, and $a + bi$ form for non-real numbers. No notes or calculator unless your teacher says so. **Total: 100 pts**

Keep in Mind

This is a practice test. It mirrors the real Unit 2 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Reading & Graphing Parabolas (20 pts)

1. (7 pts) The graph of a quadratic function f is shown. Read each feature from the graph.



Vertex: _____
 Axis of symmetry: _____
 Maximum or minimum? _____
 Its value: _____
 Zeros: _____
 Increasing on: _____
 Range: _____

2. (3 pts) Which describes the graph of $y = -2(x - 3)^2 + 5$?

(A) opens up; vertex (3, 5)

(B) opens down; vertex (-3, 5)

(C) opens down; vertex (3, 5)

(D) opens up; vertex (-3, 5)

3. (5 pts) For $f(x) = 2x^2 + 8x + 3$, use $x = -\frac{b}{2a}$ to find the axis of symmetry and the vertex.

Axis: _____
 Vertex: _____

4. (5 pts) For $y = (x + 4)(x - 2)$, find the zeros, the axis of symmetry, and the vertex.

Zeros: _____
 Axis: _____
 Vertex: _____

Part B — Solving by Factoring (20 pts)

Factor completely: take out the GCF first. To solve, get the equation equal to zero, factor, and set each factor equal to zero.

5. (8 pts) Factor completely.

(a) $3x^2 - 12x - 36$

(b) $x^2 - 81$

(c) $x^2 - 10x + 25$

(d) $3x^2 + 4x - 4$ (by grouping)

6. (4 pts) Solve $x^2 + 3x - 28 = 0$.

7. (4 pts) Solve $2x^2 = 10x$.

Solution: _____

Solution: _____

8. (4 pts) Solve $(x + 2)(x - 3) = 6$.

Solution: _____

Part C — Square Roots & Complex Numbers (20 pts)

Get the square alone before you take the root, and keep the $\pm$. Write non-real numbers in $a + bi$ form.

9. (3 pts) Solve $3x^2 + 5 = 65$.

10. (4 pts) Solve $(x - 3)^2 = 12$.

Solution: _____

Solution: _____

11. (4 pts) (a) Simplify $\sqrt{-72}$: _____ (b) Solve $x^2 + 49 = 0$: _____

12. (6 pts) Let $z_1 = 4 + 3i$ and $z_2 = 2 - 5i$. Write each answer in $a + bi$ form.

(a) $z_1 + z_2 =$ _____

(b) $z_1 - z_2 =$ _____

(c) $z_1 \cdot z_2$

$z_1 \cdot z_2 =$ _____

13. (3 pts) $\sqrt{-4} \cdot \sqrt{-25} =$ (A) 10 (B) -10 (C) $10i$ (D) $-10i$

Part D — Completing the Square, Formula & Discriminant (20 pts)

14. (4 pts) Solve by completing the square: $x^2 + 8x - 5 = 0$

Solution: _____

For 15 and 16, use the quadratic formula. Write the equation in standard form first.

15. (5 pts) Solve $2x^2 + 1 = 6x$.

16. (5 pts) Solve $x^2 + 6x + 13 = 0$.

Solution: _____

Solution: _____

17. (6 pts) For each equation, find the discriminant $b^2 - 4ac$. Then give the number and type of solutions (*two real*, *one repeated real*, or *two complex*) and the number of x -intercepts of its graph.

Equation	Discriminant	Solutions	x -intercepts
$x^2 - 6x + 9 = 0$	_____	_____	_____
$2x^2 + 3x + 4 = 0$	_____	_____	_____
$x^2 + 5x - 2 = 0$	_____	_____	_____

Part E — Systems & Modeling (20 pts)

Give each solution of a system as an ordered pair. Answer each model question with units.

18. (7 pts) Solve the system by substitution.

$$y = x^2 - 4x + 1 \quad y = x - 3$$

19. (3 pts) How many points do the graphs of $y = x^2 + 1$ and $y = 2x$ have in common?

- (A) none
(B) exactly one
(C) exactly two
(D) infinitely many

Solutions: _____

20. (10 pts) At the egg drop, an egg is launched straight up from the roof of the science building. Its height, in feet, t seconds after launch is

$$h(t) = -16t^2 + 48t + 64.$$

- (a) (2 pts) How high is the roof? _____

- (b) (4 pts) When does the egg reach its greatest height, and what is that height?

- (c) (4 pts) When does the egg hit the ground? Which solution do you reject, and why?

When: _____ Height: _____ Lands at: _____

Rejected: _____